

7(a)	arrow pointing vertically down the page	B1
7(b)	$E = \frac{1}{2}mv^2$	C1
	$E = 460 \times 1.60 \times 10^{-19} \text{ (= } 7.36 \times 10^{-17} \text{ (J))}$	C1
	$v = [(2 \times 460 \times 1.60 \times 10^{-19}) / (9.11 \times 10^{-31})]^{1/2}$ $= 1.3 \times 10^7 \text{ ms}^{-1}$	A1
7(c)	β^- particles have range of/different/various speeds/velocities/momenta/energies	M1
	so they follow different paths	A1